

Performance driven Character Effects in “The Garfield Movie”

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Figure 1: PFX in Production: Facial Expressions (left), Animation Pass without PFX (middle), Lighting Render after PFX Pass (right) .

ABSTRACT

The stylised animation in the “The Garfield Movie” presented a number of technical challenges especially for the fur which included detached limbs sliding over the surface of the body which required the groom to maintain its styling and interact reliably and predictably. Additionally limbs and body regions could be stretched drastically to support the extreme poses and stylised facial expressions requiring the fur to dynamically maintain density in those regions.

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1 INTRODUCTION

Garfield’s iconic facial features and body stripes are the hallmark of his design. In the comic strips, Jim Davis artfully manipulates these stripes, shifting and concealing them in response to expressions and poses. Playfully exaggerating silhouettes and poses, he mimics

motion blur using multiple limbs and body features. Embracing Davis’ playful spirit, characters were endowed with detachable limbs and stripes that slid across the body surface, echoing the lively chaos of the 2D realm. Translating 2D fluidity into structured 3D poses posed a unique challenge, demanding a disciplined approach. Resolving fur to hide junctions between limbs and the body, understanding stripe interactions with body fur and predicting the behavioral response of the fur to extreme poses required careful consideration. Realizing the need for a post-animation process we introduced a new pipeline task between Animation and traditional CFX to automate this performance-driven process.

2 THE SYSTEM

PFX is an automated, art-directable, and scalable process tailored to handle performance-driven effects without requiring artist intervention at the shot level. Setups are crafted and maintained based on character assets, empowering the performance team to trigger multiple shots seamlessly. Its functionalities include:

- Aligning fur orientation on intersecting and/or sliding bind meshes
- Preserving fur volumes by removing intersections
- Enhancing facial performances with cleaner deformation
- Revealing hair on stretched areas

This batchable, automated process operates in Houdini without relying on solvers. It adheres to character templates, allowing it to be executed and triggered through a proprietary batching tool on a multi-shot basis.

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3 THE CHALLENGE

During our exploration, we pinpointed three essential areas to address to uphold the essence of Davis' art while ensuring production manageability.

3.1 Detachable Limbs & Animated Stripes

To preserve the essence of 2D animation, we opted to detach the character's limbs from the body form, enabling dynamic posing with limbs gracefully sliding across the body's surface. The sliding iconic stripes added complexity, requiring seamless integration with the body fur as they traversed the surface.

Initially exploring a rigging solution for fur orientation at the limb-body junction and stripes, proved challenging without compromising rig performance or incurring additional processing costs. This challenge sparked the idea of performance-friendly solutions and significant potential for automation through an additional Houdini-driven pipeline task. After exploration, we established a stable process leveraging velocity-based curve deformation on a chunkable frame basis.

3.2 Dynamically Revealing Fur On Stretched Areas

In modern CG pipelines, exaggerated character performances cause mesh stretching, conflicting with static follicle placement and resulting in undesirable low-density patches in shots. The stretch difference between bind and performance meshes can extend well beyond 8 times the original surface size, pulling apart static hair follicles and exposing the underlying skin.

To address this, we introduced an additional process into PFX, calculating the delta between bind and performance meshes to dynamically reveal additional hair density in stretched regions, mitigating undesired effects of static follicle placement during exaggerated performances.

3.3 Automation

The challenge in automating and ensuring process autonomy presented multiple hurdles:

- **Large Number of Shots and Characters:** Managing a considerable volume of shots and characters posed a significant challenge.
- **Fast-Paced Deliveries:** The need for swift deliveries added complexity, demanding an efficient and speedy process.
- **Stability in a Dynamic Pipeline:** Maintaining stability within an ever-changing pipeline posed challenges with continuous updates and modifications.
- **Reviewing & QC Processes:** Establishing effective reviewing and quality control processes became crucial for maintaining desired standards.

Our solution involved implementing a batchable process, integrated around a proprietary bot and individual dispatching tools. This setup allowed parallel processes, meeting diverse requirements across various global sites. The core process was built on a master template with multiple character definitions, each containing character-specific overrides tailored to unique asset needs. The process invoked a predefined logic, determining which character

should be referenced by the shot in the master template. For user interaction we aimed to simplify complexity resulting in a plain interface with a shot selector for reliable use by any production team member.

4 PFX OUTPUTS & FURBALL INTEGRATION

PFX consists of a cohesive set of per-character flexible subtasks. It produces a collection of deformed guide curves, skin meshes, and map passes, complete with dynamically generated attributes. These outputs serve as driving forces for hair or deformation properties, including variables like visibility.

DNEG's proprietary fur software, "Furball", provides highly adaptable inputs and outputs, applicable at a character, template, or show level. This flexibility played a crucial role in facilitating a smooth and scalable integration of PFX's output into the fur-procedural. During the integration process, the retrieval of the latest character information from the master template logic was a key step. This information, once retrieved, was then linked to the procedural through DNEG's internal data management system. The implementation of automated syncing rules was crucial to guarantee a synchronized deployment of all dependencies associated with each PFX package across all sites, especially in the lead-up to rendering events.

5 PUSHING FOR AUTOMATION

The automation of the process introduced significant workflow efficiencies, enabling the generation of a larger number of finalized shots with minimal human intervention. This, in turn, freed up valuable resources, allowing teams to focus more on hero work. The performance teams also experienced notable advantages, such as accelerated turnarounds and dynamic feedback provided them with increased visibility, enabling quicker addressing of notes in the production process.

6 CONCLUSION AND FUTURE WORK

The PFX task consistently achieved final approvals. CFX could seamlessly overlay PFX on a per-element basis, precisely targeting adjustments only where needed. The system's flexibility empowered the team to address groom stylization challenges more efficiently, fostering a connection between our groom and performance teams. Looking ahead, substantial development potential exists in terms of automation and pipeline integration. This includes potential PDG network utilization, delegating the trigger to production or a review status, and expanding achievable tasks. Guide-based communication with the fur-procedural, dynamic decimation, and scatter are among the avenues for further exploration.

To streamline coordination and management, PFX integration into the CFX workflow was decided upon. Despite multiple stakeholders, the process is overseen by a single team. Triggered by the performance team, CFX manages both method and approval processes.

In conclusion, PFX not only provided automated performance-driven processes but also allowed targeted addressing of individual character notes as overrides. It facilitated the connection between assets and elements, crucial for advanced in-shot character performances, and contributed to a robust reviewing process.